

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai Accredited

by NAAC with “A” Grade | Accredited by NBA T1

NAGAPATTINAM–611002



B.E. Biomedical Engineering

Second Year – Fourth Semester

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302BM401	Bio Sensors and Measurements	PCC	3	0	0	3	40	60	100
2302BM402	Communication Engineering for Biomedical Engineers	PCC	3	0	0	3	40	60	100
2302BM403	Biomedical Signals & Systems	PCC	3	1	0	4	40	60	100
2302BM404	Diagnostic and Therapeutic Equipment I	PCC	3	0	2	3	40	60	100
2302BM405	Biomedical Instrumentation	PCC	3	0	2	3	40	60	100
2301HSX01	Universal Human values and Ethics	BSC	1	0	2	2	40	60	100
2301MC402	Mandatory Course -1(Circular Economy)	MC	3	0	0	0	100	0	100
PRACTICAL COURSES									
2302BM451	Biosensors and Measurements Laboratory	PCC	0	0	2	1	60	40	100
2302BM452	Diagnostic and Therapeutic Equipment Laboratory	PCC	0	0	2	1	60	40	100
2302BM453	Biomedical Instrumentation Laboratory	PCC	0	0	2	1	60	40	100
2304GE401	Professional Development Course 1I	EEC	0	0	2	1	100	--	100
TOTAL			19	1	14	22	620	480	1100

2302BM401		BIOSENSORS AND MEASUREMENTS	L	T	P	C
		(B.E- BME)	3	0	0	3

PREREQUISITE : Electron devices, Biomedical circuits and networks

COURSE OBJECTIVES:

	1. To Understand the Units and Standards of measurements for various physical quantities and how to use the measurement for calibration and error analysis.
	2. To Explain the Characteristics of the Transducer using their models and Responses.
	3. To make an experiment on various Resistance type transducer using their principle of operation and Applications.
	4. To gain knowledge about Bio Sensors and their Applications.
	5. To acquire the knowledge of another special Transducer.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1	Explain the Science of Measurement and Error Analysis
CO2	Identify the characteristics of transducers and its Responses
CO3	Experiment with Variable Resistance Transducers and their Applications.
CO4	Describe the working function of Different types of Bio Sensors and their applications
CO5	Explain the working principles of special Transducers.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	2	1	1	1		1		1
CO2	3	2	2	2	2	1	1		1		1
CO3	3	2	2	1	2	1	1		1		1
CO4	3	2	2	2	2	1	1		1		1
CO5	3	2	2	2	3	1	1		1		1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1		1
CO2	1		1
CO3	1		1
CO4	1		1
CO5	1		1

COURSE CONTENTS

Module I	SCIENCE OF MEASUREMENT	9 Hours
Measurement system – Instrumentation - Units and Standards - calibration methods, statics calibration. classification of errors- error analysis, statistical methods - odds and uncertainty		
Module II	CHARACTERISTICS OF TRANSDUCERS	9 Hours
Static characteristics - accuracy, precision, sensitivity, linearity. mathematical model of transducers – zero, first order and second - order transducers - response to impulse step, ramp and sinusoidal inputs		
Module III	VARIABLE RESISTANCE TRANSDUCERS	9 Hours
Resistance Potentiometer - Principle of operation, construction details, characteristics and applications - strain gauges- resistance thermometers- thermistors- hot-wire anemometer and humidity sensors.		
Module IV	BIOSENSORS - PHYSIOLOGICAL RECEPTORS	9 Hours
Type of Bio Sensor - Chemoreceptors, Baroreceptors, sensors for smell, sound, vision and taste. Biosensors - Working Principle and Applications		
Module V	SPECIAL TRANSDUCERS	9 Hours

Piezoelectric transducers, magnetostrictive transducers, IC sensor digital transducers - smart sensor - fibre optic transducers-Introduction to MEMS and Nano Sensors	
TOTAL	45 Hours
TEXTBOOK	
1. L. A Geddes and L.E.Baker , „Principles of Applied Biomedical Instrumentation“ Third Edition, – John Wiley and sons, Reprint 2008.	
2.Albert D.Helfrick and William D.Cooper.“Modern Electronic Instrumentation and Measurement Techniques”, Prentice Hall of India, 2007.	
REFERENCES:	
1.S.M. Sze, “Semiconductor Sensors,” New York, 1994, John Wiley & Sons.	
2.L. Ristic, “Sensor Technology and Devices,” 1994, Artech House, Inc.	
3. <u>John G. Webster</u> , <u>HalitEren</u> “Measurement, Instrumentation, and Sensors Handbook: Electromagnetic, Optical, Radiation, Chemical, and Biomedical Measurement”, 2017	
4. Jacob Fraden, “Handbook of Modern Sensors: Physics, Designs, and Applications”, Fourth Edition, Springer	

2302BM402		COMMUNICATION ENGINEERING FOR BIOMEDICAL ENGINEERS	L	T	P	C
			3	0	0	3
PREREQUISITE :						
	1.Semiconductor physics					
	2.Biomedical circuits and networks					
	3.Biomedical signals and systems					
COURSE OBJECTIVES:						
	1. Explain the concepts of generation and detection of amplitude modulation schemes					
	2. Describe the concepts of generation and detection of angle modulation schemes					
	3.Explain the concepts of multiplexing and secured communication Schemes					
	4.Explain the concepts of generation and detection of amplitude modulation schemes					
	5.Illustrate the usage of communication modalities to communicate physiological signals					
COURSE OUTCOMES:						
On the successful completion of the course, students will be able to						
CO1	1.Describe the mathematical techniques of generation, transmission and reception of amplitude modulation (AM).					
CO2	2.Explain the concepts of frequency modulation and its bandwidth					
CO3	3. Illustrate the transmission and reception of signals using digital modulation schemes					
CO4	4. Classify the usage of PN Sequence for efficient transmission of signal communication					
CO5	5. Discuss about the multiple access technique used in signal communication					
COs Vs POs MAPPING:						

	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
	CO1	3	2	1							1	1	
	CO2	3	2	1							1	1	
	CO3	3	2	1							1	1	
	CO4	3	2	1							1	1	
	CO5	3	2	1							1	1	
COs Vs PSOs MAPPING													
					COs	PSO 1	PSO 2	PSO 3					
					CO1		2	1					
					CO2		2	1					
					CO3		2	1					
					CO4		2	1					
					CO5		2	1					
COURSE CONTENTS:													
Module 1		AMPLITUDE MODULATION										9 Hours	
Introduction to communication system-Need for modulation - Amplitude modulation -Signals and Spectral Analysis of AM, DSB-SC, SSB & VSB-signal-to- noise ratio (SNR)-Receivers for continuous wave modulation - Super heterodyne Receivers.													
Module 1I		ANGLE MODULATION										9 Hours	

Basic concepts of frequency modulation .single tone frequency modulation, spectrum Analysis of sinusoidal FM wave -Narrow band FM -Wide band FM, Constant Average power Transmission band width of FM wave-Noise.		
Module III	DIGITAL MODULATION	9 Hours
Amplitude Shift Keying (ASK) – Frequency Shift Keying (FSK) Minimum Shift Keying (MSK) –Phase Shift Keying (PSK) – BPSK – QPSK - Quadrature Amplitude Modulation (QAM)		
Module IV	SPREAD SPECTRUM MODULATION	9 Hours
PN sequences – A notion of spread spectrum –Direct Sequence Spread Spectrum with coherent binary phase shift keying –Signal space dimensionality and Processing gain– Frequency Hop Spread Spectrum		
Module V	MULTUSER RADIO COMMUNICATION	9 Hours
Introduction-Multiple access technique-Satellite communication-Radio link analysis-Wireless communication-CDMA and TDMA wireless communication system		
TOTAL		45 Hours
LEARNING MATERIALS:		
1 Bernard Sklar, “Digital Communication, Fundamental and Application” Pearson Education Asia, 2nd Edition, 2001		
2.Wayne Tomasi, “Advanced Electronic Communication Systems”, Pearson Education, 6 th edition,		
3.Simon Haykin, “Communication Systems”, John Wiley & Sons, 4 th edition,2004		
4.H.Taub, D L Schilling and G Saha, “Principles of Communication”, Pearson Education, 3 rd edition,2007.		
5.B. P.Lathi, “Modern Analog and Digital Communication Systems”, Oxford University Press, 3 rd edition, 2007.		
6.R.S.Khandpur, ‘Handbook of Bio-Medical instrumentation’, Tata McGraw Hill Publishing Co Ltd.,3 rd edition, 2014.		

2302BM403	BIOMEDICAL SIGNALS AND SYSTEMS	L	T	P	C
		3	1	0	4

PREREQUEST:

Linear Algebra, Calculus and Partial Differentiation, Calculus, Ordinary Differential Equations And Complex Variables

COURSE OBJECTIVES:

	1. To study and analyze the continuous and discrete-time signals and systems, their properties and representations.
	2. To have Knowledge of time-domain representation and analysis concepts as they relate to difference equations, impulse response and convolution, etc.
	3. To familiarize the concepts of frequency-domain representation and analysis using Fourier Analysis tools, Z-transform.
	4. To understand the concepts of the sampling process and to identify and solve engineering problems
	5. To analyze the systems by examining their input and output signals

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1	Analyze the properties of signals & systems
CO2	Apply Laplace transform, Fourier transform in signal analysis
CO3	Analyze continuous time LTI systems using Fourier and Laplace Transforms
CO4	Apply Z transform and DTFT in signal analysis for Discrete time signals
CO5	Analyze discrete time LTI systems using Z transform.

COs Vs POs MAPPING:

CO#	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
1	3	2	-	-	-	-	-	-	-	1	1
2	3	2	1	1	-	-	-	-	-	1	1
3	3	2	1	-	1	-	-	-	-	1	1
4	3	2	1	-	-	-	-	-	-	1	1
5	2	1	-	-	-	-	-	-	-	1	1

COs Vs PSOs MAPPING

CO#	PSO1	PSO2	PSO3
1	2	1	-
2	2	1	-
3	2	1	-
4	3	2	-
5	3	3	-

Module I	CLASSIFICATION OF SIGNALS AND SYSTEMS	9+3 Hours
Classification of Signals - Continuous time signals - Discrete time signals - Periodic and Aperiodic signals - Even and odd signals - Energy and power signals -Deterministic and random signals - Complex exponential and Sinusoidal signals. Classification of Systems: Continuous time systems - Discrete time systems - Linear system - Time Invariant system – causal system - BIBO system - Systems with and without memory - LTI system Classification of Systems.		
Module II	ANALYSIS OF CONTINUOUS TIME SIGNALS	9+3 Hours

Fourier series analysis -Trigonometric Fourier series, Cosine Fourier series, Exponential Fourier series, Fourier Spectrum of continuous time signals, Fourier transform analysis, Laplace transform, Analysis of electrical network using Laplace transform.		
Module III	LTI CT SYSTEM	9+3 Hours
Analysis of differential equation -Transfer function-Impulse response - Frequency response - Convolution integral ,Fourier Methods - Laplace transforms analysis - Block diagram representation - State variable equation and Matrix.		
Module IV	ANALYSIS OF DISCRETE TIME SIGNALS	9+3 Hours
Spectrum of DT signals - Discrete Time Fourier Transform (DTFT) - Properties of discrete time Fourier transform - Discrete Fourier Transform (DFT) - Properties of DFT - Z-transform in signal analysis - Properties of Z - transform - Inverse Z – transform		
Module V	LTI DISCRETE TIME SYSTEMS	9+3 Hours
Analysis of differential equation - Transfer function - Impulse response - Convolution sum - Analysis and characterization of DT system using Z transform Difference Equations-Block diagram.		
		Total: 45+15 Hours
Further Reading:		
Programs using mathematical computing tool for CT and DT system analysis using LT and ZT		
References:		
1. Allan V.Oppenheim, S.Wilsky and S.H.Nawab, “Signals and Systems”, Pearson, 2007.		
2. B. P. Lathi, “Principles of Linear Systems and Signals”, Second Edition, Oxford, 2009.		
3. R.E.Zeimer, W.H.Tranter and R.D.Fannin, “Signals & Systems - Continuous and Discrete”, Pearson, 2007.		
4. John Alan Stuller, “An Introduction to Signals and Systems”, Thomson, 2007.		
5. Hwei. P.Hsu, Schaum’s Outlines: Signals and Systems, Pearson Education, 2002.		
6. Anand Kumar A, “Signals and Systems” , PHI learning Pvt. Ltd., Second edition, 2012.		
7. Simon Haykin and Barry Van Veen, “Signals and Systems” , John Willey & Sons, Inc., Second edition, 2004.		

2302BM404		DIAGNOSTIC AND THERAPEUTIC EQUIPMENT - I					L	T	P	C			
		(B.E-BME)					3	0	2	3			
PREREQUISITE:													
		1. Biosciences in Medical Engineering											
		2. Human Anatomy and Physiology											
COURSE OBJECTIVES:													
		1. Understand the medical equipment used in the measurement of parameters related to cardiology											
		2. Discuss the equipment used in neurology											
		3. Demonstrate EMG recording unit and its uses											
		4. Explain diagnostic and therapeutic devices related to respiratory parameters.											
		5. Understand the various sensory measurements that hold clinical importance.											
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1	Describe the working and recording setup of all essential cardiac equipment.												
CO2	Understand the working and recording of all essential neurological equipment.												
CO3	Discuss about muscular and biomechanical measurements												
CO4	Explain about respiratory measurement system.												
CO5	Describe the measurement techniques of sensory responses.												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	
	CO1	3	2	1						1		1	
	CO2	3	2	1						1		1	
	CO3	3	2	1	1					1		1	
	CO4	3	2	1						1		1	
	CO5	3	2	1		1				1		1	
COs Vs PSOs MAPPING													
COURSE CONTENTS:													
Module I		CARDIAC EQUIPMENT									9 Hours		
Electrocardiograph, Normal and Abnormal Waves, Heart rate monitor, Holter Monitor, Phonocardiography, ECG machine maintenance and troubleshooting, Cardiac Pacemaker Internal and External Pacemaker– Batteries, AC and DC Defibrillator													
Module II		NEUROLOGICAL EQUIPMENT									9 Hours		
Clinical significance of EEG, Multi-channel EEG recording system, Epilepsy, Evoked Potential– Visual, Auditory and Somatosensory, MEG (Magneto Encephalo Graph). EEG Bio Feedback Instrumentation. EEG system maintenance and troubleshooting.													
Module III		MUSCULAR AND BIOMECHANICAL MEASUREMENTS									9 Hours		
Recording and analysis of EMG waveforms, fatigue characteristics, Muscle stimulators, nerve stimulators, Nerve conduction velocity measurement, EMG Bio Feedback Instrumentation. Static Measurement – Load Cell, Pedobarograph.													

Module IV	RESPIRATORY MEASUREMENT SYSTEM	9 Hours
Instrumentation for measuring the mechanics of breathing – Spirometer -Lung Volume and vital capacity, measurements of residual volume, Pneumotachometer – Airway resistance measurement, Apnoea Monitor. Types of Ventilators – Pressure, Volume, and Time controlled. Flow, Patient Cycle Ventilators, Humidifiers, Nebulizers, Inhalators.		
Module V	SENSORY MEASUREMENT	9 Hours
Psychophysiological Measurements – polygraph, basal skin resistance (BSR), galvanic skin resistance (GSR), Sensory responses - Audiometer-Pure tone, Speech, Eye Tonometer, Application Tonometer, slit lamp, auto refractometer.		
TOTAL:		45 Hours
Text books:		
1. John G. Webster, —Medical Instrumentation Application and Design, 4th edition, Wiley India Pvt.Ltd, New Delhi, 2015.		
2. Joseph J. Carr and John M. Brown, —Introduction to Biomedical Equipment Technology, Pearson education, 2012.		
References:		
1. Khandpur R.S, “Handbook of Biomedical Instrumentation”, Tata McGraw Hill, New Delhi, 2003..		
2. Leslie Cromwell, “Biomedical Instrumentation and Measurement”, Prentice Hall of India, New Delhi, 2007		
3. John G. Webster, “Medical Instrumentation Application and Design”, John Willey and Sons, 2006.		
4. Joseph J. Carr and John M. Brown, “Introduction to Biomedical Equipment Technology”, Pearson Education, 2004.		
5. Richard Aston “Principles of Biomedical Instrumentation and Measurement”, Merril Publishing Company, 1990.		
6. L.A Geddas and L.E.Baker “Principles of Applied Biomedical Instrumentation” 2004.		
7. John G. Webster, “Bioinstrumentation”, John Willey and sons, New York, 2004.		

2302BM405	BIOMEDICAL INSTRUMENTATION	L	T	P	C
		3	0	2	3

PREREQUISITE:

	Human Anatomy and Physiology, Electron devices and electrical circuits and networks
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COURSE OBJECTIVES:

1. To illustrate origin of bio potentials and its propagations to understand the different types of electrodes and its placement for various recordings.
2. To design bio amplifier for various physiological recordings and analyze different measurement techniques for non-physiological parameters
3. To Summarize different biochemical measurements

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Categorize different bio potential electrodes based on its origin, propagations and characteristics.
CO2:	Apply different bio-potential measurements techniques and analyze the characteristics of bio signals
CO3:	Apply various Biosignal conditioning techniques as a pre-processing method in Biosignal processing.
CO4:	Apply various technique for non-electrical physiological measurements
CO5:	Analyze the performance of biochemical sensors and biochemical measurement techniques

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1							1
CO2	3	2	1	1							1
CO3	3	2	1	1							1
CO4	3	2	1	1							1
CO5	3	2	1	1							1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1	-	
CO2	1	-	
CO3	1	-	2
CO4	1	-	2
CO5	1	-	2

COURSE CONTENTS:

Module I	BIOPOTENTIAL ELECTRODES	09 Hours
Origin of biopotential and its propagation, Electrode-electrolyte interface, electrode-skin interface, half-cell potential, Contact impedance, polarization effects of electrode-non polarizable electrodes, Types of electrodes surface, needle, and microelectrodes and their equivalent circuits. Recording problems- motion artifacts, measurement with two electrodes.		
Module II	BIOPOTENTIAL MEASUREMENTS	09 Hours
Bio signals characteristics-frequency and amplitude ranges. ECG-Einthoven's triangle, standard 12 lead system, Principles of vector cardiography. EEG-10-20 electrode system, unipolar, bipolar and average mode. EMG-unipolar and bipolar mode. Recording of ERG, EOG and ECG.		
Module III	BIOSIGNAL CONDITIONING	09 Hours
Need for bio-amplifier-single ended bio-amplifier, differential bio-amplifier, impedance matching circuit, isolation amplifiers- transformer and optical isolation- isolated DC amplifier and AC carrier amplifier, power line interference, Right leg driven ECG amplifier, Band pass filtering		
Module IV	MEASUREMENT OF NON- ELECTRICAL PARAMETERS	09 Hours
Temperature, respiratory rate and pulse rate measurements. Blood Pressure: indirect methods- Auscultatory method, oscillometric method, direct methods: electric manometer, pressure amplifiers, systolic, diastolic, mean detector circuit. Blood flow and cardiac output measurement: Indicator dilution, thermal dilution and dye dilution method, Electromagnetic and ultrasound blood flow measurement.		
Module V	BIOCHEMICAL MEASUREMENT AND BIOSENSORS	09 Hours
Biochemical sensors- pH, pO ₂ and pCO ₂ , Ion selective Field Effect Transistor (ISFET), Immunologically sensitive FET(IMFET), Blood glucose sensors, Blood gas analyzers- colorimeter, Sodium Potassium Analyser, spectrophotometer, blood cell counter, auto analyzer (simplified schematic description)- Biosensors-Principles-amperometric and voltometric techniques, Electrophoretic techniques.		
TOTAL: 45 Hours		
Text books:		
1. John G. Webster, —Medical Instrumentation Application and Design, 4th edition, Wiley India Pvt.Ltd, New Delhi, 2015.		
2. Khandpur R.S, “Handbook of Biomedical Instrumentation”, Tata McGraw Hill, New Delhi, 2014.		
References:		
1. Joseph J. Carr and John M. Brown, —Introduction to Biomedical Equipment Technology, Pearson education, 2012.		
2. 16. Leslie Cromwell, “Biomedical Instrumentation and Measurement”, Prentice Hall of India, New Delhi, 2007		
3. 17. L.A Geddes and L.E.Baker, “Principles of Applied Biomedical Instrumentation”, John Wiley and Sons, 3rd Edition, Reprint 2008.		
4. 18. Myer Kutz, “Standard Handbook of Biomedical Engineering & Design”, McGraw-Hill Publisher, 2003.		

2301HSX01	UNIVERSAL HUMAN VALUES AND ETHICS	L	T	P	C
		1	0	2	2

PREREQUISITE:

1. Ethical thinking

COURSE OBJECTIVES:

1. Reinstate India's rich cultural legacy and human values of which we are the custodians.
2. Focus on professional ethics, which help citizens to discern desirable and undesirable actions.
3. Re-emphasize constitutional values, universal values, and holistic education to create integrated citizens.
4. Lay down broader guidelines of human values and ethics for internal and external stakeholders.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:** Apply critical thinking skills to solve problems and make informed decisions in various contexts.
- CO2:** Analyze the principles of effective self - governance and evaluate their implementation in different scenarios.
- CO3:** Understand the importance of a fair and transparent system of rewards and reprimand in constitution of India and apply these principles in real-world situations.
- CO4:** Analyze the role of an individual to develop social reliability and create awareness about their assertiveness and self - confidence.
- CO5:** Understand the knowledge of inner qualities and instruments of one's self in stress management and analyze the effects of meditation in one's physical, mental, social and spiritual well - being.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	2	-	-	1	1	3	1	-	1
CO2	2	1	2	-	-	1	2	2	1	-	1
CO3	2	1	2	-	-	1	1	2	1	-	1
CO4	2	1	2	-	-	1	1	2	1	-	1
CO5	2	1	2	-	-	1	2	3	1	2	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-

COURSE CONTENTS:

Module I	INTRODUCTION TO INDIAN ETHOS	08 Hours
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Meaning of ethos and cultural essence of India – Scriptures as the base of the Indian Knowledge System (IKS) – Integrating the two methodologies: interiorization process for self-exploration, and exterior scientific pursuit for the prosperity of world – The Law of Karma and Nishkama Karma (The Law of action and selfless action).

Practical: Five hours of Yoga practice per week, Ethics through Music and Indian Poetry, Community Engagement.

Module II	HUMAN VALUES AND ETHICS	09 Hours
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Knowing the Self and the universal values that we stand for - This is self enquiry & self discovery – Background conversations and deep listening - recognizing the assumptions that we make - the biases we have - and the implications for ethical action – Self-identity: distinguishing and embracing oneself (and others) four profiles (inner-potential, social, professional, personality) – Distinguish ideology, perspectives beliefs from embodying values.

Practical: Self discovery, self enquiry and Mindfulness, Yama & Niyama of Ashtang Yoga.

Module III	CONSTITUTIONAL VALUES AND GLOBAL CITIZENSHIP	09 Hours
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Values embedded in the Preamble of the Indian Constitution Integration of Human Rights and duties – Directive principles and responsibilities as citizens of India – Sensibility and responsibilities towards global environment, Loksangraha and Vasudhaiva Kutumbakam.

Practical: Debates and Theatre on diversity and plurality, research on similarities and differences in the ethos of different countries.

Module IV	VALUES AND SKILLS FOR YOUTH	09 Hours
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Designing to make a difference through strategies using the Conscious Full Spectrum Response model – Listening for commitment behind complaints to transform contentious arguments and create a space for listening and change – Distinguishing judgement from discernment – Being assertive and confident (assertiveness incorporates self-confidence).

Practical : Development of concentration among students through music, fine arts, mathematics, sports, yoga and mindfulness

Module V	INTEGRATED PERSONALITY AND WELL-BEING	10 Hours
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The three gunas (qualities of sattva—purity and harmony, rajas —activity and passion, tamas — darkness and chaos), the four antah-karanas (inner instruments), and panchkosha (five sheaths) – Stress management: meditated personality and agitated personality – Oneness, non-duality, and equanimity – Physical, mental, social, and spiritual well-being.

Practical : Talks on importance of the Ayurvedic concept of well being and nutrition, sports activities

TOTAL: 45 Hours

TEXT BOOKS:

1. R R Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.

REFERENCES:

1. Blanchard, Kenneth and Peale, Norman Vincent. 1988. *The Power of Ethical Management*. New York: William Morrow and Company, Inc.

2. Gandhi, Mohandas Karamchand. 1971. *Pathway to God* compiled by MS Deshpande. Ahmedabad: Navajivan Mudranalaya, Navjivan Trust.

3. <https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php>

2302BM451	BIOSENSORS AND MEASUREMENTS LABORATORY									L	T	P	C
										0	0	2	1
PREREQUEST: Biosensors and measurements, Semiconductor devices													
COURSE OBJECTIVE:													
1. To understand the principles and working mechanisms of various sensors and transducers used for measurement of physical quantities													
2. To familiarize students with the characteristics and applications of displacement, temperature, pressure, and strain measuring devices.													
3. To provide hands-on experience in operating test and measurement instruments such as CRO and DSO.													
4. To develop the ability to select suitable sensors for specific engineering applications.													
COURSE OUTCOMES:													
After completion of the course, Student will be able to													
CO1:	Explain the principles and working mechanisms of common transducers and sensors such as capacitive, pressure, strain gauge, thermocouple, LVDT, RTD, and thermistor.												
CO2:	Analyze the static and dynamic characteristics of temperature, displacement, and pressure measuring devices.												
CO3:	Operate and interpret readings from laboratory instrumentation such as CRO and DSO.												
CO4:	Select suitable sensors/transducers for given measurement applications based on their characteristics.												
CO5:	Perform experimental procedures to obtain, record, and interpret measurement data from various sensors												
COs Vs POs MAPPING:													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11		
CO1	3	2	2	1	1						1		
CO2	3	2	2	1	1						1		
CO3	3	2	2	1	1						1		
CO4	3	2	2	1	1						1		
CO5	3	2	2	1	1						1		
COs Vs PSOs MAPPING													
						COs	PSO1	PSO2	PSO2				
						CO1	1	1	-				
						CO2	1	1	-				
						CO3	1	1	-				
						CO4	1	1	-				
						CO5	1	1	-				
LIST OF EXPERIMENTS													
1. To study principle and working of a capacitive Transducer.													
2. To study principle and working of a pressure sensor													
3. To study principle and working of a strain gage sensor.													
4. To study the principle and working of a thermocouple													
5. To study principle and working of LVDT													
6. To study the characteristics of resistance temperature detector													
7. Study of Front panel of CRO&DSO													
8. To study the characteristics of a thermistor													
Total :30 Hours													
References:													
1. Principles of applied Biomedical Instrumentation by La Geddes and L.E. Baker.													
2. Biomedical Instrumentation and Measurement by Leslie Cromwell, Fred. J. Weibell and Pfeiffer													
3. Principles of Biomedical Instrumentation and Measurement, Richard Aston, Merril Publishing Co., Columbus, 1990													

2302BM452	DIAGNOSTIC AND THERAPEUTIC EQUIPMENT LABORATORY	L	T	P	C
		0	0	2	1

PREREQUEST:

Human anatomy and physiology, Semiconductor Physics, Biomedical instrumentation

COURSE OBJECTIVES:

1. To demonstrate recording and analysis of different Bio potentials
2. To examine different therapeutic modalities.
3. To understand the process of biotelemetry.
4. To study about extracorporeal process.
5. To measure various physiological signals

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1:	1. Demonstrate the therapeutic applications of different Diathermy devices.
CO2:	2. Measure different bioelectrical signals using various methods.
CO3:	3. Explain the fundamental principles, components, and types of biomedical telemetry systems.
CO4:	4. Analyze the different bio signals using suitable tools.
CO5:	5. Examine the electrical safety measurements.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	1	-	-	-	-	-	-	1
CO2	3	3	2	-	-	-	-	-	-	-	1
CO3	3	2	2	-	2	-	-	-	-	-	1
CO4	3	2	2	-	-	-	-	-	-	-	1
CO5	2	2	1	-	2	-	-	-	-	-	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO2
CO1	3	-	2
CO2	3	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-

LIST OF EXPERIMENTS:

1. Demonstration of shortwave diathermy principles, operation.
2. Demonstration of ultrasonic diathermy and its application in physiotherapy
3. Measurement of Respiratory parameters using spirometry
4. Study of hemodialysis model
5. Galvanic skin resistance (GSR) measurement
6. Measurement of visually evoked potential
7. Measurement of various physiological signals using biotelemetry
8. Study of medical stimulator
9. Measurement of Electrical safety

Total :30 Hours

References:

1. Webster, J. G., & Nimunkar, A. J. (2021). Medical instrumentation 5ed: Application and design (Indian adaptation) (pp. 656). Wiley India. ISBN-10 9354249159.
2. McPherson, R. A., & Pincus, M. R. (2021). Henry's clinical diagnosis and management by

laboratory methods (24th ed.). Elsevier.
3. Cromwell, L., Weibell, F. J., & Pfeiffer, E. A. (2015). Biomedical instrumentation and measurement (2nd ed.). New Delhi: Prentice Hall of India.
4. Joseph J. Carr and John M. Brown, (2012). Introduction to biomedical equipment technology. Pearson Education.

2302BM453	BIOMEDICAL INSTRUMENTATION LABORATORY	L	T	P	C
		0	0	2	1

PREREQUEST:

Human physiology

COURSE OBJECTIVES:

1. To provide hands on training on Measurement of physiological parameters.
2. To study about blood flow and blood measurement
3. To understand about patient monitoring system

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1:	Measure the hydrogen ion concentration in a solution
CO2:	Determine the electrical activity of heart, brain and muscles
CO3:	Record and analyze pulse rate and respiration rate.
CO4:	Measure the blood flow velocity by ultrasound transducer.
CO5:	Measure blood pressure and blood flow

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1							1	1
CO2	3	2	1	1							1	1
CO3	3	2	1	1							1	1
CO4	3	2	1	1							1	1
CO5	3	2	1	1							1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO2
CO1	2	1	-
CO2	2	1	-
CO3	2	1	-
CO4	2	1	-
CO5	2	1	-

LIST OF EXPERIMENTS:

1. Measurement of Hydrogen ion using pH meter
2. Recording of ECG signal and analysis using Simulation
3. Simulation and analysis of EMG amplifier.
4. Simulation and analysis of EEG with Simulation
5. Measurement of respiration rate.
6. Measurement of blood flow velocity using ultrasound transducer.
7. Measurement of blood pressure using sphygmomanometer.
8. Study of Baby ventilator

Total :30 Hours

References:

1. Medical Instrumentation – Application and Design” by John G Webster
2. “Transducers for Biomedical Measurements: Principles and Applications” by Richard S C Cobbold
3. Measurement Systems, Application and Design” by Ernest O Doebelin

2301MC402	CIRCULAR ECONOMY	L	T	P	C
		2	1	2	0
COURSE OBJECTIVES:					
1.	To develop graduates who have the necessary theoretical, practical and research knowledge, skill and aptitude in circularity and can get job opportunities by the industry in various sectors both public and private at national and international level.				
2.	To contrive skilled manpower and entrepreneurship in the field of Circular Economy.				
3.	To enhance interaction of students with the senior/experienced manpower who have real time knowledge / experience in the technology development, research, innovation, entrepreneurship deployment and circular business models.				
4.	To acquaint students about the needs of businesses related to circularity and to create zeal among students to pursue research and development (R&D), and Entrepreneurship in this domain.				
5.	Create entrepreneurs who would promote knowledge in core competencies of environmental education and work on “innovation to industry” approach through university-industry partnerships.				
COURSE OUTCOMES:					
Upon successful completion of the course, students will be able to					
CO1	Apply the concept of circular economy to environmental engineering problems				
CO2	Understand the concept of circularity and conduct relevant research				
CO3	Use the principles of circularity for application to sustainable development				
CO4	Apply complexity aspects of circular economy for creating circular business				
COURSE CONTENTS:					
Module-I	INTRODUCTION TO CIRCULAR ECONOMY				9 Hours
Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy					
Module-II	CHARACTERISTICS OF CIRCULAR ECONOMY				9 Hours
Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops					
Module-III	CIRCULAR DESIGN, INNOVATION AND ASSESSMENT				9 Hours
Zero waste: Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business Models					
Module-IV	CASE STUDIES				9 Hours
Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks					
Module-V	LEGAL AND POLICY FRAMEWORK				9 Hours
Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy, ESG					
TOTAL: 45 HOURS					
TEXTBOOKS:					
1. The Circular Economy A User's Guide, Walter R Stahel Routledge; 1st Edition (24 June 2019)					

2. Circular Economy: (Re) Emerging Movement, Shalini Goyal Bhalla Invincible Publisher
3. The Circular Economy Handbook: Realizing The Circular Advantage, Peter Lacy, Jessica Long, Wesley Spindler Palgrave Macmillan UK
4. Waste to Wealth: The Circular Economy Advantage, Peter Lacy, Jakob Rutqvist Palgrave Macmillan
REFERENCES:
1. Towards Zero Waste: Circular Economy Boost, Waste to Resources María-Laura Franco-García, Jorge Carlos Carpio-Aguilar, Hans Bressers. Springer International Publishing 2019
2. Strategic Management and the Circular Economy Marcello Tonelli, Nicolo Cristoni, Routledge 2018.
3. Circular Economy: Global Perspective Sadhan Kumar Ghosh, Springer, 2020
4. The Circular Economy: A User's Guide Stahel, Walter R. Routledge 2019
5. An Introduction to Circular Economy Lerwen Liu, Seeram Ramakrishna, Springer Singapore 2021.